Lathi 20题练习册 第8批

对应 Boulet 前四章

原书英文题干、公式与必要图示 · 无答案版

记号：D = d/dt；差分式中 E 为前移算子，E{y[n]} = y[n+1]；j^2 = -1
u 为单位阶跃，δ 为单位冲激；LTIC / LTID 分别指连续 / 离散时间 LTI 系统

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来源：B. P. Lathi / R. A. Green, Linear Systems and Signals, 3rd ed. (2018)
中文主题仅用于课程对应；题目要求以原书英文为准。

半占空比周期锯齿脉冲的能量与功率

Lathi 1.1-7 | 原书第 137、138 页

Boulet 第1章：Periodic Signals， p.8；Finite-Energy and Finite-Power Signals， p.21

1.1-7 Figure P1.1-7 shows a periodic 50% duty cycle dc-offset sawtooth wave $x(t)$ with peak amplitude A. Determine the energy and power of $x(t)$.

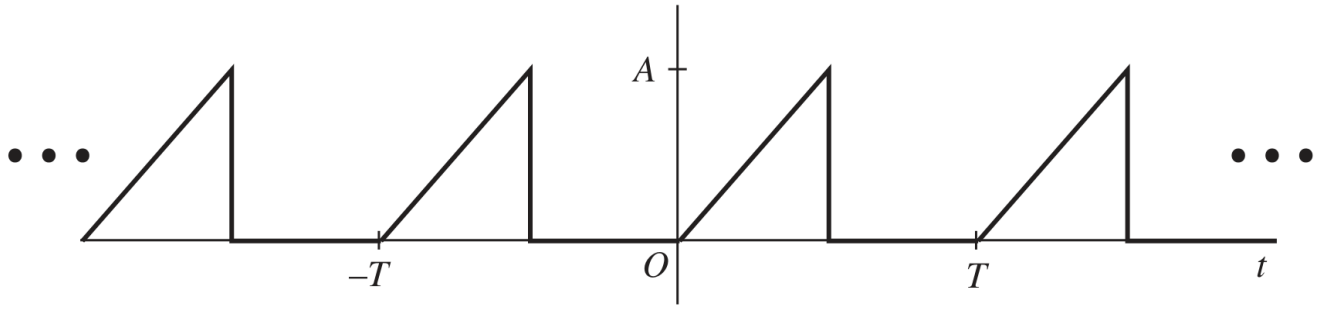


Figure P1.1-7

演算区

分段线性波形的平移、反转与伸缩

Lathi 1.2-2 | 原书第 138、139 页

Boulet 第1章：Transformations of the Time Variable, p.4

1.2-2 For the signal $x(t)$ illustrated in Fig. P1.2-2, sketch

(a) $x(t-4)$
(b) $x(t/1.5)$
(c) $x(-t)$
(d) $x(2t-4)$
(e) $x(2-t)$

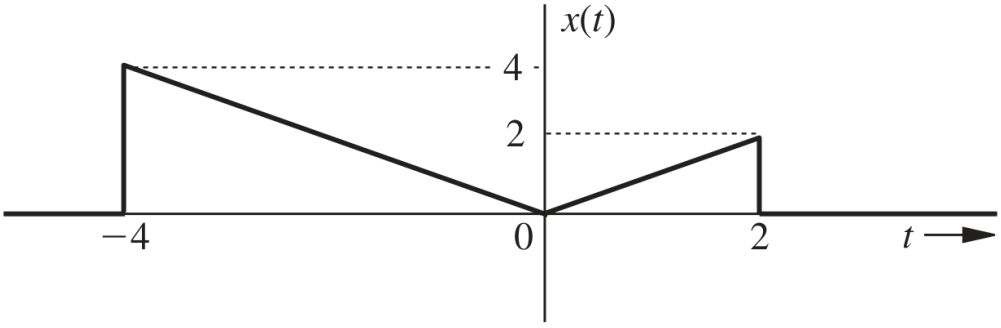


Figure P1.2-2

演算区

分段线性信号的广义一阶与二阶导数

Lathi 1.4-8 | 原书第 142、139、141 页

Boulet 第1章：Generalized Functions，p.26

所需图P1.2-2和图P1.4-2(a)均已附，分别用于(a)、(b)。

1.4-8 (a) Find and sketch dx/dt for the signal $x(t)$ shown in Fig. P1.2-2.
(b) Find and sketch d^2x/dt^2 for the signal $x_1(t)$ depicted in Fig. P1.4-2a.

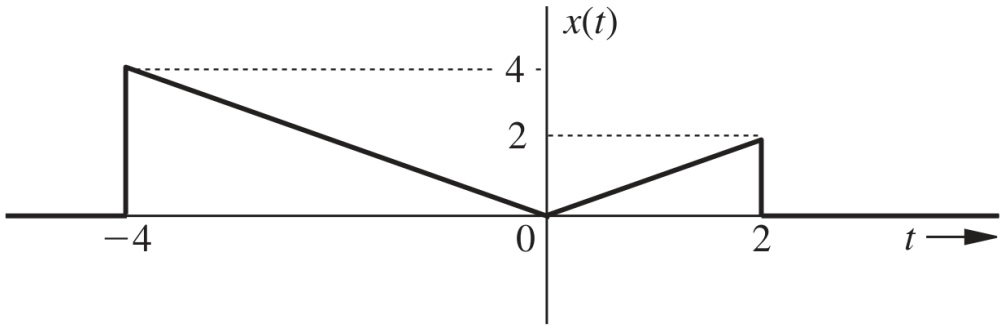
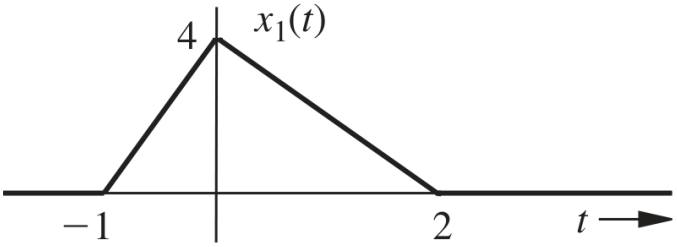


Figure P1.2-2



(a)

演算区

奇偶分量的正交性与积分性质

Lathi 1.5-4 | 原书第 143 页

Boulet 第1章：Even and Odd Signals, p.23

1.5-4 (a) If $x_e(t)$ and $x_o(t)$ are even and the odd components of a real signal $x(t)$, then show that

$$\int_{-\infty}^{\infty} x_e(t)x_o(t)\,dt = 0$$

(b) Show that

$$\int_{-\infty}^{\infty} x(t)\,dt = \int_{-\infty}^{\infty} x_e(t)\,dt$$

演算区

周期离散序列变换后的能量与功率

Lathi 3.2-5 | 原书第 316 页

Boulet 第1章：Transformations of the Time Variable，p.4；Periodic Signals，p.8；Finite-Energy and Finite-Power Signals，p.21

保留原书n=0箭头。(b)下标为2-n/3；按原书离散时间扩展约定，下标非整数的位置补零。

3.2-5 Letting $\downarrow$ identify the $n = 0$ value, describe a 4-periodic signal $w[n]$ using vector notation as $[\cdots, 1, 2, 3, 4, \overset{\downarrow}{1}, 2, 3, 4, 1, 2, 3, 4, \cdots]$.

(a) Determine the energy E_x and power P_x of signal $x[n] = w[2n]$.

(b) Determine the energy E_y and power P_y of signal $y[n] = w[2 - \frac{n}{3}]$.

演算区

阶跃冲激响应下分段脉冲的图解卷积

Lathi 2.4-4 | 原书第 225、226 页

Boulet 第2章：Continuous-Time LTI Systems: The Convolution Integral，p.67

所需原图P2.4-4已附，保留负脉冲幅值与全部时间刻度。

2.4-4 An LTIC system has impulse response $h(t) = 3u(t-2)$. For input $x(t)$ shown in Fig. P2.4-4, use the graphical convolution procedure to determine $y_{zsr}(t) = h(t) * x(t)$. Accurately sketch $y_{zsr}(t)$. When solving for $y_{zsr}(t)$, flip and shift $x(t)$ and explicitly show all integration steps—even if apparently trivial!

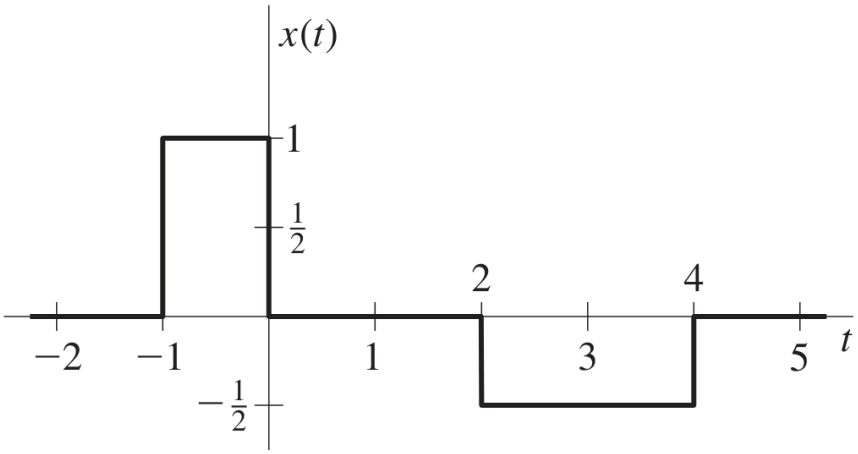


Figure P2.4-4

演算区

阶跃输入与分段线性冲激响应的卷积

Lathi 2.4-30 | 原书第 228、230 页

Boulet 第2章：Continuous-Time LTI Systems: The Convolution Integral，p.67

2.4-30 Determine $y(t) = x(t) * h(t)$ for the signals depicted in Fig. P2.4-30.

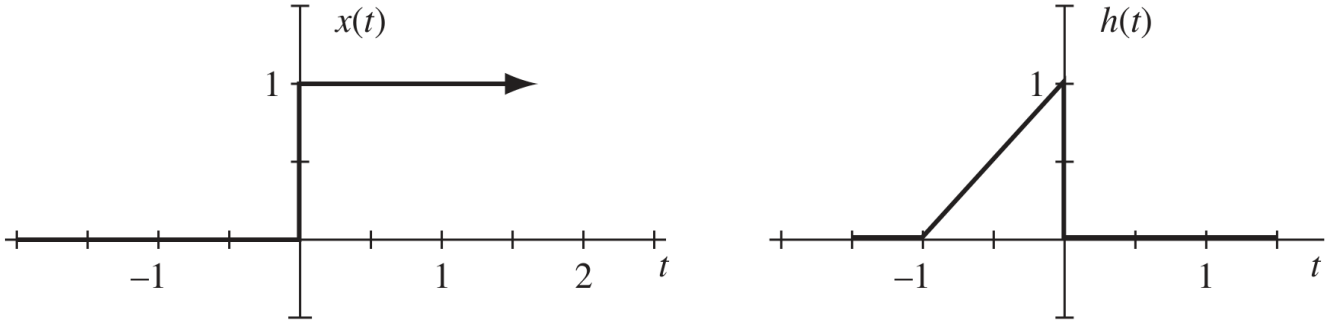


Figure P2.4-30

演算区

连续LTI系统并联与串联的等效冲激响应

Lathi 2.4-33 | 原书第 230 页

Boulet 第2章：Properties of Linear Time-Invariant Systems， p.74

题干(a)-(c)与原图P2.4-33(a)、(b)均完整附上。

2.4-33 Two LTIC systems have impulse response functions given by $h_1(t) = (1-t)[u(t) - u(t-1)]$ and $h_2(t) = t[u(t+2) - u(t-2)]$.

(a) Carefully sketch the functions $h_1(t)$ and $h_2(t)$.

(b) Assume that the two systems are connected in parallel, as shown in Fig. P2.4-33a. Carefully plot the equivalent impulse response function, $h_p(t)$.

(c) Assume that the two systems are connected in cascade, as shown in Fig. P2.4-33b. Carefully plot the equivalent impulse response function, $h_s(t)$.

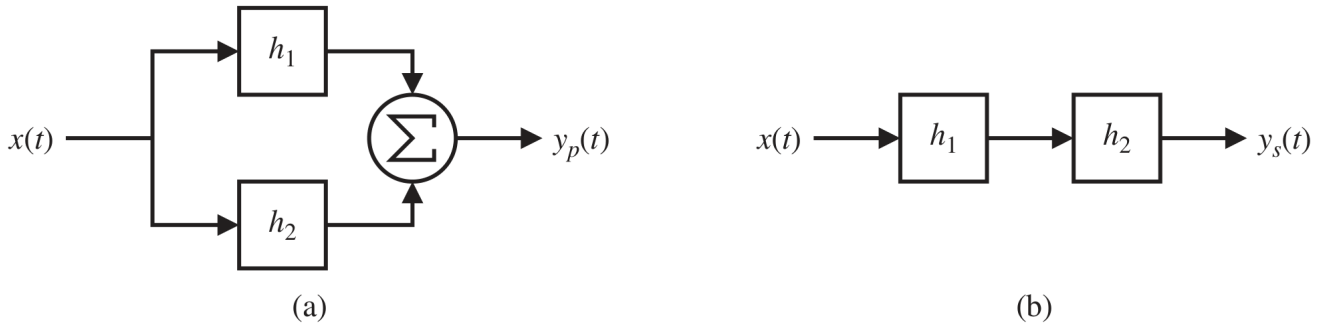


Figure P2.4-33

演算区

带时移的离散指数卷积：求和与查表对照

Lathi 3.8-5 | 原书第 322、285 页

Boulet 第2章：Discrete-Time LTI Systems: The Convolution Sum，p.54

原题要求分别用卷积和及表3.1求响应；附表3.1表头和本题所需第4行作为已知参考公式，未代入本题参数求解。

3.8-5 Find the (zero-state) response $y[n]$ of an LTID system whose unit impulse response is

$$h[n] = (-2)^n u[n-1]$$

and the input is $x[n] = e^{-n}u[n+1]$. Find your answer by computing the convolution sum and also by using Table 3.1.

TABLE 3.1 Select Convolution Sums

No.	$x_1[n]$	$x_2[n]$	$x_1[n] * x_2[n] = x_2[n] * x_1[n]$	
4	$\gamma_1^n u[n]$	$\gamma_2^n u[n]$	$\left[\frac{\gamma_1^{n+1} - \gamma_2^{n+1}}{\gamma_1 - \gamma_2}\right] u[n]$	$\gamma_1 \neq \gamma_2$

演算区

非递归离散系统的阶跃响应与初值需求

Lathi 3.8-17 | 原书第 322 页

Boulet 第2章：Discrete-Time LTI Systems: The Convolution Sum， p.54；Properties of Linear Time-Invariant Systems， p.74

3.8-17 For a system specified by equation

$$y[n] = x[n] - 2x[n-1]$$

Find the system response to input $x[n] = u[n]$. What is the order of the system? What type of system (recursive or nonrecursive) is this? Is the knowledge of initial condition(s) necessary to find the system response? Explain.

演算区

衰减振荡模态的连续系统零输入响应

Lathi 2.2-8 | 原书第 224 页

Boulet 第3章：Causal LTI Systems Described by Differential Equations，p.92；Characteristic Polynomials and Stability，p.112

展开所引用2.2-4的要求：(a)求特征多项式、特征方程、特征根和特征模态；(b)按给定初值求 t>=0 的零输入响应。初始导数15.98按原书保留。

2.2-8 Repeat Prob. 2.2-4 for

$$(D^2+4D+13)y(t)=4(D+2)x(t)$$

with $y_0(0^-)=5,\ \dot{y}_0(0^-)=15.98.$

演算区

含重复零特征根系统的初值响应

Lathi 2.2-9 | 原书第 224 页

Boulet 第3章：Causal LTI Systems Described by Differential Equations，p.92；Characteristic Polynomials and Stability，p.112

展开所引用2.2-4的要求：(a)求特征多项式、特征方程、特征根和特征模态；(b)按给定三个初值求 t>=0 的零输入响应。

2.2-9 Repeat Prob. 2.2-4 for

$$D^2(D+1)y(t) = (D^2+2)x(t)$$

with $y_0(0^-) = 4$, $\dot{y}_0(0^-) = 3$, and $\ddot{y}_0(0^-) = -1$.

演算区

五点移动平均的差分模型与延迟实现

Lathi 3.4-3 | 原书第 317 页

Boulet 第3章：Causal LTI Systems Described by Difference Equations， p.96

3.4-3 A moving average is used to detect a trend of a rapidly fluctuating variable, such as the stock market average. A variable may fluctuate (up and down) daily, masking its long-term (secular) trend. We can discern the long-term trend by smoothing or averaging the past N values of the variable. For the stock market average, we may consider a 5-day moving average $y[n]$ to be the mean of the past 5 days' market closing values $x[n], x[n-1], \ldots, x[n-4]$.

(a) Write the difference equation relating $y[n]$ to the input $x[n]$.

(b) Use time-delay elements to realize the 5-day moving-average filter.

演算区

由位置采样估计速度、加速度及冲激响应

Lathi 3.4-16 | 原书第 319 页

Boulet 第3章：Causal LTI Systems Described by Difference Equations，p.96；Impulse Response of a Difference LTI System，p.109

原题同页跨栏，采样率、物理导数关系和(a)、(b)均完整附上。

3.4-16 A jet-powered car is filmed using a camera operating at 60 frames per second. Let variable n designate the film frame, where $n = 0$ corresponds to engine ignition (film before ignition is discarded). By analyzing each frame of the film, it is possible to determine the car position $x[n]$, measured in meters, from the original starting position $x[0] = 0$.

From physics, we know that velocity is the time derivative of position:

$$v(t) = \frac{d}{dt}x(t)$$

Furthermore, we know that acceleration is the time derivative of velocity:

$$a(t) = \frac{d}{dt}v(t)$$

We can estimate the car velocity from the film data by using a simple difference equation $v[n] = k(x[n] - x[n-1])$.

(a) Determine the appropriate constant k to ensure $v[n]$ has units of meters per second.

(b) Determine a standard-form constant coefficient difference equation that outputs an estimate of acceleration, $a[n]$, using an input of position, $x[n]$. Identify the advantages and shortcomings of estimating acceleration $a(t)$ with $a[n]$. What is the impulse response $h[n]$ for this system?

电阻梯形网络的二阶差分边值问题

Lathi 3.6-7 | 原书第 320、318 页

Boulet 第3章：Causal LTI Systems Described by Difference Equations，p.96；相关差分方程求解练习

所需原图P3.4-8及提示所引差分方程已附；节点编号n=0,1,...,N。本题边界条件在两端给定，并非通常的相邻时间初值。

3.6-7 Find $v[n]$, the voltage at the nth node of the resistive ladder depicted in Fig. P3.4-8, if $V = 100$ volts and $a = 2$. [*Hint* 1: Consider the node equation at the nth node with voltage $v[n]$. *Hint* 2: See Prob. 3.4-8 for the equation for $v[n]$. The auxiliary conditions are $v[0] = 100$ and $v[N] = 0$.]

$$v[n+2] - Av[n+1] + v[n] = 0 \quad A = 2 + \frac{1}{a}$$

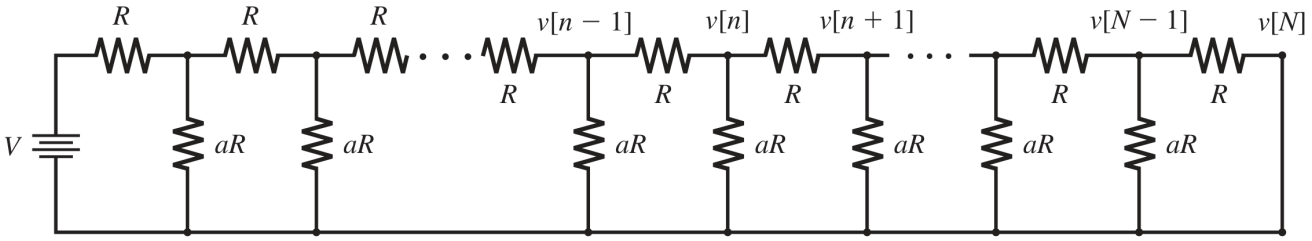


Figure P3.4-8

演算区

（复习）谐波、基频与三角傅里叶级数系数

Lathi 6.1-5 | 原书第 669 页

Boulet 第4章：Linear Combinations of Harmonically Related Complex Exponentials，p.132；Determination of the Fourier Series Representation，p.134

复习题：此前见第1批。复习第1批：本题无需积分，可集中检查基频识别、谐波编号与三角级数系数，为后续谱分析打好基础。已重新核对原书，(a)至(f)全部保留；(c)(d)为复值信号，三角级数系数也可以是复数。

6.1-5 Determine the trigonometric Fourier series coefficients a_n and b_n for the following signals. In each case, also determine the signals' fundamental radian frequency ω_0. No integration is required to solve this problem.

(a) $x_{\mathrm{a}}(t) = \cos(3\pi t)$

(b) $x_{\mathrm{b}}(t) = \sin(7\pi t)$

(c) $x_{\mathrm{c}}(t) = 2 + 4\cos(3\pi t) - 2j\sin(7\pi t)$

(d) $x_{\mathrm{d}}(t) = (1+j)\sin(3\pi t) + (2-j)\cos(7\pi t)$

(e) $x_{\mathrm{e}}(t) = \sin(3\pi t + 1) + 2\cos(7\pi t - 2)$

(f) $x_{\mathrm{f}}(t) = \sin(6\pi t) + 2\cos(14\pi t)$

演算区

（复习）分段周期信号的复指数傅里叶级数

Lathi 6.3-2 | 原书第 672 页

Boulet 第4章：Determination of the Fourier Series Representation of a Continuous-Time Periodic Signal，p.134

复习题：此前见第1批。复习第1批：重练按一周期分段积分求复指数系数，尤其区分直流项与非零谐波。已重新核对原书，跨栏的题干、分段函数及要求证明的Dn公式均完整保留；所附Dn表达式本来就是原题的证明目标，不是另附解答。

6.3-2 A 2π-periodic signal $x(t)$ is specified over one period as

$$x(t) = \begin{cases} \dfrac{1}{A}t & 0 \le t < A \\ 1 & A \le t < \pi \\ 0 & \pi \le t < 2\pi \end{cases}$$

Sketch $x(t)$ over two periods from $t = 0$ to 4π. Show that the exponential Fourier series coefficients D_n for this series are given by

$$D_n = \begin{cases} \dfrac{2\pi - A}{4\pi} & n = 0 \\ \dfrac{1}{2\pi n}\left(\dfrac{e^{-jAn} - 1}{An} + je^{-jn\pi}\right) & n \neq 0 \end{cases}$$

演算区

全波整流正弦波的Parseval功率与谐波截断项数

Lathi 6.3-13 | 原书第 675、631 页

Boulet 第4章：Optimality and Convergence of the Fourier Series，p.144；Parseval Theorem，p.150

本题为新题。完整题干、引用的图6.17及Drill6.5给出的已知指数傅里叶级数均已附。图示信号为x(t)=|sin t|，周期为π；所附已知级数的系数是Dn=2/[π(1-4n^2)]。本题明确保留从n=-N至N的2N+1项，要求这些项的功率不低于原信号的99.75%；未附所求N或功率计算的答案。

6.3-13 The signal $x(t)$ in Fig. 6.17 is approximated by the first $2N+1$ terms (from $n=-N$ to N) in its exponential Fourier series given in Drill 6.5. Determine the value of N if this $(2N+1)$-term Fourier series power is to be no less than 99.75% of the power of $x(t)$.

$$x(t)=\frac{2}{\pi}\sum_{n=-\infty}^{\infty}\frac{1}{1-4n^2}e^{j2nt}$$

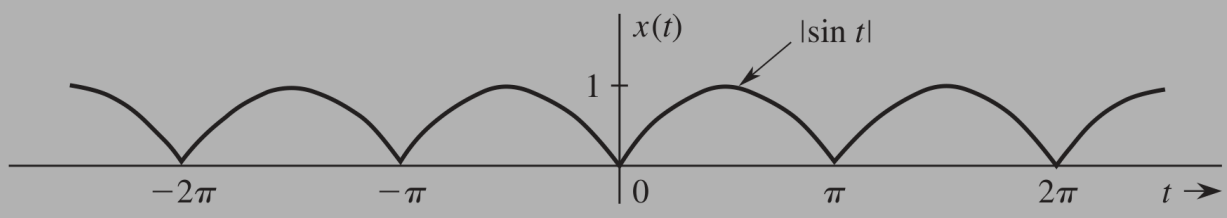


Figure 6.17 Full-wave rectified sine wave for Drill 6.5.

演算区

（复习）周期抛物线波经过理想带通滤波器后的稳态输出

Lathi 6.4-3 | 原书第 675 页

Boulet 第4章：Determination of the Fourier Series Representation，p.134；Steady-State Response of an LTI System to a Periodic Signal，p.155

复习题：此前见第1批。复习第1批：把周期波形的系数计算与LTI系统对各谐波的筛选连起来，覆盖Boulet第4章周期稳态响应。已重新核对完整分段定义及(a)至(c)。原题带通滤波器的通带宽度为1 Hz、中心频率为3 Hz，留意Hz与角频率单位的区别。

6.4-3 Consider a $T_0 = 1$ periodic signal $x(t)$ defined as

$$x(t) = \begin{cases} 1 - t^2 & 0 < t \leq 1 \\ x(t+1) & \forall t \end{cases}$$

(a) Sketch $x(t)$ for $-2 \leq t \leq 2$.
(b) Determine D_n, the exponential Fourier series spectrum of $x(t)$.
(c) If $x(t)$ is applied to an ideal bandpass filter with a 1 Hz passband centered at 3 Hz, determine the output $y(t)$.

演算区

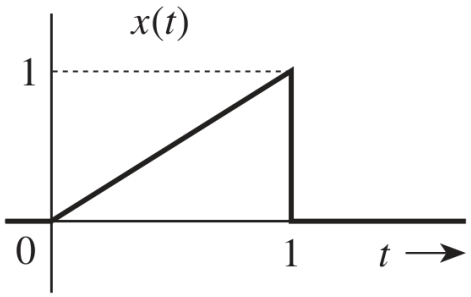
（复习）斜坡信号的有限项傅里叶逼近与误差能量

Lathi 6.5-6 | 原书第 677 页

Boulet 第4章：Optimality and Convergence of the Fourier Series，p.144；Parseval Theorem，p.150

复习题：此前见第1批。复习第1批：重练三角傅里叶级数的截断逼近，用误差能量衡量有限项表示的效果，与Boulet的最优性和Parseval内容直接对应。已重新核对原书并附所需图P6.5-4(a)；只使用该图(a)，不使用图(b)。本题在0<t<1上表示x(t)=t，指定基频为2π；原题first N terms未另行说明直流项是否计入，作答时请写明项数约定。

6.5-6 Represent the signal $x(t)$ shown in Fig. P6.5-4a over the interval from 0 to 1 by a trigonometric Fourier series of fundamental frequency $\omega_0 = 2\pi$. Compute the error energy in the representation of $x(t)$ by only the first N terms of this series for $N = 1, 2, 3$, and 4.



(a)

Figure P6.5-4

演算区